



Ophthalmic Image Synthesis and Analysis with Generative Adversarial Network Artificial Intelligence

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Abstract

Generative adversarial networks (GANs), introduced by Ian Goodfellow in 2014, have revolutionized adversarial machine learning, particularly in data synthesis. This manuscript explores their application in ophthalmic diagnostics, addressing the scarcity of annotated datasets and the need for improved early disease detection. By leveraging GAN architectures, the goal is to enhance the quality of synthetic ophthalmic images, ultimately improving diagnostic algorithm training. A systematic review was conducted from January to April 2024 across PubMed, Embase, and Scopus. Search terms included “Generative Adversarial Networks” and “ophthalmic image synthesis.” Articles were selected based on relevance to retinal image generation and diagnostic improvement in ophthalmology. GANs show considerable promise in generating high-resolution retinal and optical coherence tomography (OCT) images. Models like DR-GAN and Pix2Pix have successfully synthesized images that resemble real diagnostic data, proving valuable when annotated datasets are scarce. GAN-generated images enhance training for algorithms detecting diseases such as diabetic retinopathy, glaucoma, and age-related macular degeneration. Recent advances, including conditional GANs and CycleGANs, have enabled disease-specific image generation, boosting the diversity of training datasets, particularly in resource-limited settings. Integrating GANs into ophthalmic diagnostics represents a significant leap in medical AI, offering high-quality synthetic images to improve diagnostic algorithms. Despite their potential, challenges such as the need for larger datasets, improved image interpretability, and noise reduction must be addressed. Future research should focus on optimizing these models and incorporating multi-modal data to enhance diagnostic accuracy in clinical settings.

Keywords Image synthesis · Generative adversarial networks · GAN · OCT · Diffusion models

Introduction

We present a comprehensive framework that synthesizes and categorizes the current and emerging applications of generative adversarial networks (GANs) in ophthalmology. The concept of adversarial machine learning, especially GANs, was first introduced in 2014 by Ian Goodfellow et al. in their landmark paper [1]. In this framework, two neural networks—the generator and the discriminator—compete against each other in a dynamic feedback loop. The generator aims to create synthetic data that mimics real-world samples, while the discriminator’s role is to differentiate between real and generated data [2]. Through this adversarial process, both networks progressively enhance their performance: the generator produces increasingly realistic

data, and the discriminator becomes more adept at identifying synthetic data (Figs. 1 and 2) [1].

Ophthalmology faces numerous diagnostic challenges, such as the limited availability of diverse, annotated datasets, the need for early detection of ophthalmic diseases, and the complexity of interpreting medical images [5–7]. Traditional diagnostic methods can often be subjective, time-consuming, and susceptible to inter-observer variability, underscoring the need for more standardized, efficient, and innovative solutions to improve diagnostic accuracy. GANs offer a promising solution to these challenges by generating synthetic data that can supplement limited datasets, enhance image quality, and support the development of automated diagnostic systems. Through image synthesis and data augmentation, GANs can create diverse and representative datasets that improve the generalization and robustness of machine learning models in analyzing ophthalmic images

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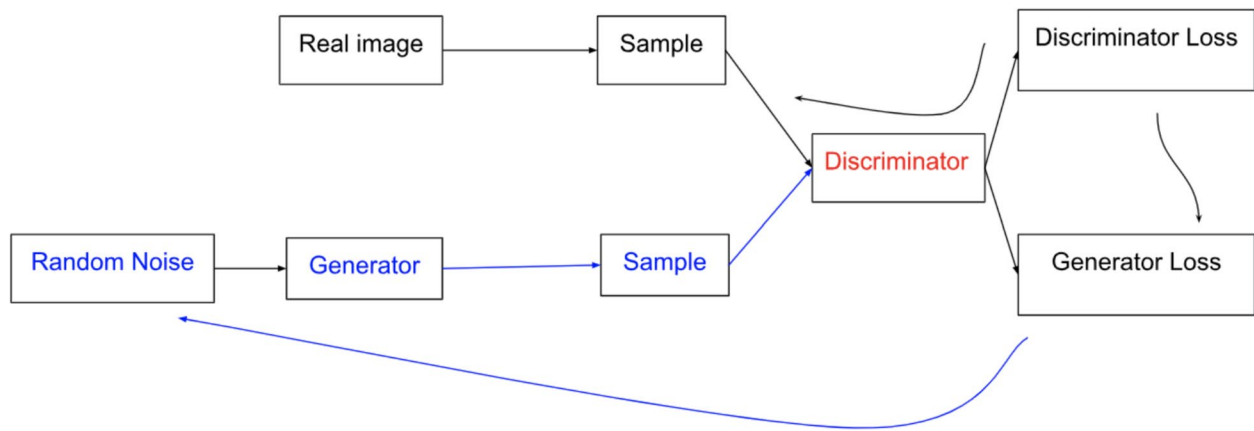


Fig. 1 An example of how a GAN operates: the generator produces synthetic images from random noise, while the discriminator has to find the difference between real and generated images. The blue

arrow represents the backpropagation of the discriminator's loss, while the red arrow indicates the backpropagation of the generator's loss [3]

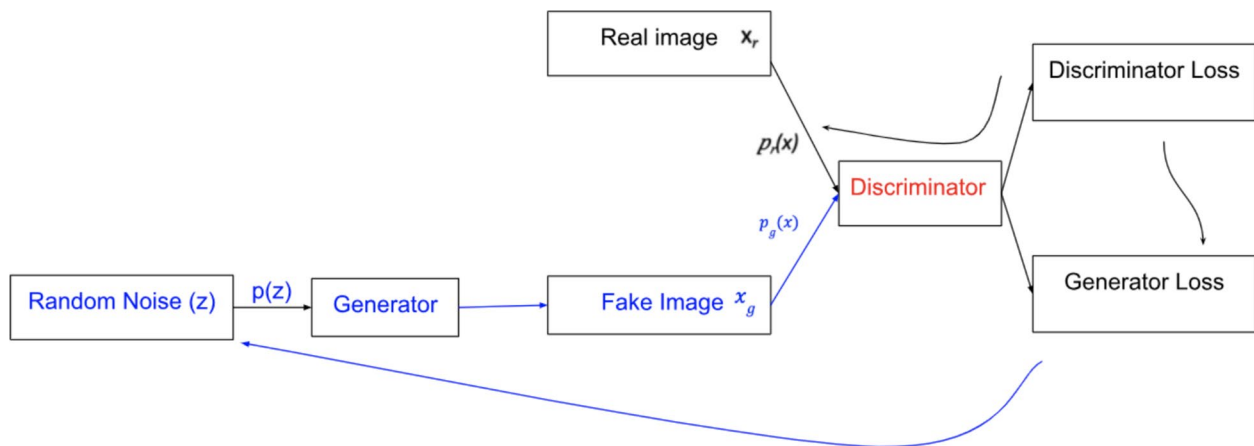


Fig. 2 An example of how a GAN can be applied for retinal image synthesis: the generator converts a noise vector z from the distribution $p(z)$ into a synthesized retinal image x_g . The discriminator then distinguishes between synthetic and real retinal images based on the

distributions of x_g and x_r , respectively. The generated image samples follow a distribution $p_g(x)$, which tries to approximate the real image distribution $P_r(x)$ [4]

[8]. Moreover, GANs can generate high-resolution retinal images [9–14], reduce noise in imaging [9, 15, 16], and enhance image clarity [17–23], ultimately improving the accuracy of disease detection and classification algorithms.

Recent surveys have highlighted the increasing application of GANs in ophthalmology. Zhang et al. addressed many of the key applications of GANs in ophthalmic image domains, categorizing studies into disease-specific synthesis, noise reduction, and data augmentation [24]. This survey found that segmentation, or assigning a pixel or areas of an image to a category, was one of the most studied tasks in ophthalmology. These primarily revolved around fundus photography of the retinal vessels, optic disc, and optic cup, as these are historically known to be challenging due to the variations in widths, colors, branching, and tortuosity [24].

Glaucoma, age-related macular degeneration (AMD), and early detection of diabetic retinopathy are some diseases where GANs have been used in developing more robust, highly realistic synthetic images to mimic disease states, aid in model training, and improve diagnostic accuracy. Adopting diffusion models is another trending topic, as demonstrated in denoising diffusion models for fundus photograph synthesis, offering improved stability and mode coverage over traditional GANs [25].

In clinical settings, particularly in remote or underserved areas where advanced medical imaging equipment may be scarce, GANs can generate synthetic images to diagnose [4, 26]. For pediatric patients who might find traditional imaging methods uncomfortable or stressful, GANs could provide a non-invasive alternative for

generating diagnostic images. In emergencies requiring immediate imaging or when imaging devices are unavailable or malfunctioning, GANs can rapidly generate synthetic images based on existing data, enabling prompt medical decision-making and treatment.

Though there is growing interest in applying GANs and associated models in complex medical imaging, there remains a lack of literature that brings together all the current knowledge in the field. Much of the available literature focuses on niche applications of GANs in ophthalmic image analysis while ignoring the broader view of the progression and potential of GANs in many fields. Given the rapid advancements that occur with AI and the lack of comprehensive literature, we aim to provide a holistic perspective to bridge these gaps and offer a systemic review and analysis of the applications, challenges, points

of concern, and future directions of GANs in the field of ophthalmology (Tables 1, 2, 3, 4, 5, 6, and 7).

Our proposed framework organizes GAN applications in ophthalmology into several functional categories:

- **Conditional GANs** and **Cycle GANs** for disease-specific fundus image synthesis
- **Dual-phase GANs**, **Tub-GANs**, and **Tub-sGANs** for healthy fundus image generation
- **Diabetic retinopathy GANs (DR-GANs)** for generating fundus images with pathological features
- **Progressive GANs (PGGANs)**, **deep convolutional GANs (DCGANs)**, and **StyleGANs** for synthetic OCT image synthesis
- **Denoising GANs (DN-GANs)** and **CycleGANs** for noise reduction and image enhancement

Table 1 Search strategy summary, including key details of the search strategy, including databases used, search terms, timeframe, inclusion/exclusion criteria, and the selection process for identifying studies on GAN applications in ophthalmic image synthesis

Items	Specification
Databases and sources	PubMed, Embase, Scopus
Search terms used (including MeSH and free text search terms and filters)	“Image Synthesis”, “Generative Adversarial Networks”, “GAN”, “OCT GAN”
Note: please use an independent supplement table to present detailed search strategy of one database as an example	
Inclusion and exclusion criteria (study type, language restrictions etc.)	Inclusion: Research Investigations, Primary Analysis Exclusion: Papers not in the English language
Selection process (who conducted the selection, whether it was conducted independently, how consensus was obtained, etc.)	M.M – Conducted study selection based on relevance to GAN applications in ophthalmic image synthesis and analysis. Consensus among the research team was obtained through discussions and meetings
Any additional considerations, if applicable	N/A

Table 2 Detailed search strategy composed of search terms, including MeSH and free-text terms, combined with Boolean operators to identify studies on GAN applications in ophthalmology, focusing on image synthesis and deep learning

Category	Terms used	Purpose
Medical Subject Headings (MeSH) Terms	"Ophthalmology", "Generative Adversarial Networks", "Image Processing, Computer-Assisted"	These standardized MeSH were used to target articles indexed with specific terms related to GANs and AI applications in ophthalmology. The terms ensure precise retrieval of articles relevant to medical imaging and AI-driven diagnostics in the ophthalmology domain
Free Text Terms	"generative AI", "deep-learning in ophthalmology", "ocular imaging applications", "retinal image synthesis", "CycleGAN", "Pix2Pix"	Free-text terms were included to capture studies that may not be comprehensively indexed under the existing MeSH terms. These terms allow for broader coverage of AI applications in ophthalmic diagnostics, particularly in areas where GANs are applied for image synthesis and analysis
Search Strategy	("Ophthalmology"[MeSH Terms]) AND ("Generative Adversarial Networks"[MeSH Terms] OR "GANs"[All Fields]) AND ("Image Processing, Computer-Assisted"[MeSH Terms] OR "deep-learning"[All Fields] OR "neural networks"[All Fields])	The search strategy combined both MeSH terms and free-text terms using Boolean operators to ensure comprehensive coverage. This approach allowed for the identification of articles focusing on ophthalmic AI applications, GAN-based image processing, and deep-learning methodologies. The combination of specific and broad terms ensured relevant studies were not missed

Table 3 Summary of key investigations on the application of GANs in fundus imaging (healthy and diseased) along with their primary findings

Authors	Methods	Key findings
Bellemo et al. [27]	Pairing retinal fundus images with vessel tree segmentation, using architectures including U-Net for mapping and generating new images. Employing a variety of loss functions to enhance the realism and quality of synthesized images	Authors highlight the successful application of GANs in improving the performance of image analysis tasks through augmented training datasets, despite the challenges associated with synthesized images that include abnormal vessel networks and the inability to accurately replicate retinal lesions
Zhao et al. [28]	Incorporating a style transfer framework to introduce diversity in the synthesized images, from a single tubular structured annotation, even with a small training set of as few as 10 examples. Effectively capturing the tubular structure while allowing for varied visual appearances	Proved the ability of this approach to work well with limited training data, its capacity to produce diverse images from the same annotation, its contribution to improving image segmentation, as well as its effectiveness in generating realistic-looking images and enhancing segmentation module performance
Chen et al. [29]	Development of Retinal Fundus Images Generative Adversarial Networks (RF-GANs) to address the imbalance in diabetic retinopathy (DR) training data for deep-learning models. Two generative models were used, RF-GAN1 for translating retinal images from one domain to another to improve data diversity, and RF-GAN2 for creating synthetic retinal images using structural and lesion masks along with DR grading labels	Successful narrowing of the domain gap between datasets by RF-GAN1, leading to improved performance of segmentation models. RF-GAN2 demonstrated the ability to synthesize realistic retinal fundus images, significantly enhancing the accuracy (+ 1.53%) and quadratic weighted kappa (+ 1.70%) of state-of-the-art DR grading models on the EyePACS testing set, showcasing the potential of RF-GANs to augment data and improve DR detection algorithms
Zhou et al. [30]	Exploring DR-GAN which involves synthesizing high-resolution fundus images conditioned on structural and lesion masks, along with adaptive grading vectors from a latent grading space. This allows for the manipulation of grading severity and the synthesis of fine details through a multiscale spatial and channel attention module which is optimized by joint adversarial losses	The model was able to generate highly realistic fundus images that significantly enhance data augmentation for diabetic retinopathy (DR) grading and lesion segmentation models. The effectiveness of DR-GAN was validated through extensive experiments on the EyePACS and FGADR datasets and demonstrated its superiority in synthesizing diverse images that improve DR grading task performance
Welander et al. [31]	The study explored the use of two unsupervised GANs, CycleGAN and UNIT, alongside two supervised models, for the image-to-image translation of T1- and T2-weighted MR images. The models were trained using a dataset from the Human Connectome Project, with images normalized for quantitative evaluation. The effectiveness of these models was measured both quantitatively (MAE, MI, PSNR) and qualitatively through a perceptual study to assess how realistic the synthesized images are	The generator model achieved the best quantitative results across most metrics, but it did not necessarily produce the most visually realistic images. Images found to be more visually realistic in the perceptual study did not always correlate with better quantitative error measurements. This discrepancy highlights the complexity of evaluating GAN-generated images based solely on traditional metrics or visual realism
Costa et al. [32]	Authors explored a GANs model which involves training an adversarial autoencoder to synthesize retinal vessel networks and a GAN to generate color retinal images from these networks, both optimized through differentiable loss functions to work jointly in an end-to-end synthesis system	The model's ability to produce anatomically consistent and visually realistic synthetic retinal images that are significantly different from the training set. The study demonstrates the model's capacity for generalization and its potential utility in training machine learning algorithms for retinal image analysis. The learned latent space exhibits a well-defined semantic structure, allowing for operations like smooth interpolation between different retinal images, thereby providing a new approach for generating synthetic medical imagery to support research and development in medical image analysis
Guibas et al. [33]	The study explores segmentation masks to capture the geometry of the dataset (Stage-I GAN) and then translate these masks into photorealistic images (Stage-II GAN). A novel two-stage pipeline utilizing GANs for generating synthetic medical images was specifically tested on retinal fundi images	F1 accuracy rating of 0.8877 for a u-net model trained on the synthetic data and 0.8988 for the model trained on real data from the DRIVE database, indicated the high quality of the synthetic training data. KL-divergence scores revealed a distinction between synthetic and real datasets' distributions, which suggests the synthetic data's unique contribution without simply replicating the original data distribution. This innovative pipeline promises to significantly enhance the availability of medical imaging data for research while maintaining patient privacy

Table 3 (continued)

Authors	Methods	Key findings
Yu et al. [34]	This study explored a novel preprocessing pipeline called multiple-channels-multiple-landmarks (MCML) along with GANs to synthesize color fundus images from a combination of vessel tree, optic disc, and optic cup images	MCML significantly outperformed single vessel-based methods against all GAN architectures tested. The Pix2pix model with the ResU-net generator demonstrated superior peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) performance compared to other GAN models. The application of high-resolution paired images further enhanced the performance of each GAN, leading to the generation of realistic fundus images
Diaz-Pinto et al. [35]	The use of Deep Convolutional Generative Adversarial Networks (DCGANs) to develop a retinal image synthesizer and a semi-supervised learning method for automatic glaucoma assessment was explored. This approach utilizes a combination of a small glaucoma-labelled database and a large unlabelled database to train a classifier capable of distinguishing between glaucomatous and normal retinal images	The model was able to generate realistic cropped retinal images and to classify them with high accuracy, achieving an Area Under the Curve (AUC) of 0.9017. This demonstrates a significant advancement in computer-aided glaucoma diagnosis

Table 4 Summary of key investigations on the application of GANs in corneal and external eye diagnostics along with their primary findings

Authors	Methods	Key findings
Jameel et al. [36]	A CycleGAN was employed to generate synthetic UBM images from SS-ASOCT images. The model's efficacy was evaluated by comparing the quality of real vs. synthetic images, assessed by glaucoma specialists, and analyzing UBM measurements of the anterior chamber and iridociliary parameters across two datasets	Authors highlight the successful application of GANs in improving the performance of image analysis tasks through augmented training datasets, despite the challenges associated with synthesized images that include abnormal vessel networks and the inability to accurately replicate retinal lesions
Yildiz et al. [37]	IVCM images from 85 subjects were collected and tested using GAN and U-Net-based segmentation methods. Pearson's R correlation, Bland–Altman analysis, and ROC statistics were used to evaluate model performance under clinical supervision by three ophthalmologists	GAN-based segmentation demonstrated superior robustness to imaging noise, particularly speckle noise, compared to U-Net. ROC analysis revealed significantly higher accuracy in GAN-based segmentation, and the ICC between graders was 0.97, reflecting a high level of agreement in manual segmentation
Zembarain et al. [38]	A two-stage deep-learning algorithm was utilized, beginning with the identification of three corneal layers in CCM images using a GAN, followed by the classification of SNP features through 21-fold cross-validation on 207 manually segmented images	The algorithm achieved high Dice similarity coefficients (background: 0.992, nerve: 0.814, neuroma: 0.748, immune cell: 0.736), indicating performance comparable to human segmentation. The strong intraclass correlation supports the clinical applicability of the method
Khan et al. [39]	A cGAN model was employed to analyze meibomian gland dysfunction (MGD) using infrared (IR) imaging, focusing on addressing challenges in quantifying gland irregularities. The model's performance was evaluated using meiboscorring, Pearson correlation, Bland–Altman analysis, average Pompeiu-Hausdorff distance, and Aggregated Jaccard Index	The method outperformed traditional techniques such as adaptive thresholding, U-Net, and GAN, showing a strong correlation ($r = 0.96, p < 0.001$) between automated and manual analyses. The ratio of MG area to total eyelid area decreased significantly as the meiboscore increases, providing an objective measure of MG loss. Intra- and inter-operative reproducibility was excellent. Artifacts from light reflection introduced inconsistencies

Table 5 Summary of the main investigations on the application of GAN in OCT imaging along with their key findings

Study	Methods	Key findings
Zheng et al. [40]	GAN's architecture was used to synthesize high-resolution OCT images using a publicly available dataset, comprising 37,206 images with choroidal neovascularization, 11,349 images with diabetic macular edema, 8617 images with drusen, and 51,140 images from normal eyes, and the synthesized images were evaluated by two retinal specialists	The image quality of real and synthetic OCT images was deemed similar by two retinal specialists with discrimination accuracy rates of 59.50% and 53.67%. The DL models trained on real and synthetic OCT images exhibited high AUC values of 0.99 and 0.98 for the local dataset and 0.94 and 0.90 for the clinical dataset
Kumar et al. [41]	GAN models were progressively trained to generate circumpapillary OCT scans and evaluated for image gradability and authenticity by two clinical experts on a set of 100 real and 100 synthetic images	In the analysis of 990 normal and 862 glaucomatous eyes, evaluations of the clinical set for gradeability and authenticity were comparable with the best-performing DL network trained on synthetic images displaying higher AUC scores on both internal (0.97) and external (0.90) test datasets compared to the network trained with real images (internal AUC: 0.96, external AUC: 0.84)
Chen et al. [42]	Application of a GAN method, dubbed DN-GAN, to reduce speckle noise for OCT images, using a frequency loss function	The approach displayed superior de-noising performance compared to state-of-the-art denoising methods with fewer parameters
Ma et al. [43]	Application of a conditional GAN (cGAN) for simultaneous speckle reduction and contrast enhancement of retinal OCT images	The proposed method outperforms both traditional and DL denoising methods, showcasing superior overall performance, good generalization capabilities, and effectiveness in despeckling various types of OCT images
Cheong et al. [44]	A GAN-based approach (i.e., DeshadowGAN) was created and trained with 2328 B-scans to eliminate blood vessel shadows in unseen B-scans	The proposed algorithm exhibited a 57.2% higher mean average gradient magnitude (AGM) compared to the current state-of-the-art methods. There were increases of 11.1% in mean peak signal-to-noise ratio (PSNR), 154% in contrast-to-noise ratio (CNR), and 187% in structural similarity index metric (SSIM) when compared to single-frame B-scans/
Cheong et al. [45]	Development of OCT-GAN to remove speckle noise and retinal shadows within a single step	The cGAN method could generate post-treatment OCT images with acceptable accuracy, specificity, and negative predictive values (95% CI range: 77.0–91.9, 94.1–95.1, and 54.7–96.5%, respectively), based on baseline OCT images. The addition of FA and ICGA images further improved these metrics
Lee et al. [46]	A conditional GAN approach (cGAN), trained on augmented paired OCT B-scan images from neovascular age-related macular degeneration patients, was evaluated on a test set to predict the presence of intraretinal fluid, subretinal fluid, pigment epithelial detachment, and subretinal hyperreflective material. The authors also compare the generated post-treatment OCT images with those generated from baseline OCT with or without fluorescein angiography (FA) and indocyanine green angiography (ICGA) images	
Liu et al. [47]	Application of GAN to generate and assess personalized post-therapeutic OCT images in predicting short-term responses to anti-VEGF therapy for typical nAMD from pretherapeutic OCT scans, involving 476 training pairs and 50 test pairs,	The study revealed that 92% of synthetic OCT images were of sufficient quality for clinical interpretation with approximately 26–30% accurately identified as synthetic. The generated post-therapeutic OCT images demonstrated a high accuracy (0.85, 95% CI 0.74 to 0.95) in predicting the macular status of wet or dry conditions
Moon et al. [48]	Development of a GAN-based model trained on 1684 OCT images from 842 patients to predict treatment outcomes associated with anti-VEGF therapy for nAMD patients	The model higher sensitivity in predicting the presence of retinal fluid after treatment compared to human examiners for both ranibizumab (sensitivity: 0.615 vs. 0.385 for examiner 1, 0.231 for examiner 2) and aflibercept (sensitivity: 0.857 vs. 0.429 for examiner 1, 0.429 for examiner 2); synthetic posttreatment images showed a fluid status difference of 18.5% between ranibizumab and aflibercept

Table 5 (continued)

Study	Methods	Key findings
Xu et al. [49]	Application of GAN to synthesize and assess post-therapeutic OCT images integrating pre-therapeutic images, using a pix2pixHD method trained on 515 pairs of images from patients with retinal vein occlusion (RVO) in the training set	The pix2pixHD algorithm accurately predicted short-term responses to anti-vascular endothelial growth factor (anti-VEGF) therapy, demonstrating comparable edema resorption in synthetic and actual OCT images. Retinal specialists had difficulty distinguishing the majority of synthetic images (62/68) from real ones
Liu et al., 2023 [50]	Application of GANs in assessing the predictive capability of OCT images in determining the response of diabetic macular edema (DME) patients to anti-VEGF therapy	OCT images generated by GAN models, particularly RegGAN, demonstrated high comparability to real images with most post-therapeutic OCT images (95/103) proving challenging for retinal specialists to differentiate from real images
Tavakkoli et al. [51]	Development of a cGAN-based approach to generate FA images from retinal fundus photographs as a non-invasive and cost-effective alternative to visualize retinal vasculature without the need for FA	The proposed cGAN model could generate anatomically accurate angiograms, comparable in fidelity to FA images, outperforming two other state-of-the-art generative algorithms; expert evaluations confirmed the model's ability to produce high-quality FA images indistinguishable from real angiograms
Ouyang et al. [21]	A cascaded neural network framework was developed, incorporating a cGAN for pre-segmentation of OCT B-scans to remove specular artifacts and speckle noise, followed by a Tissue Interface Segmentation Network (TISN) for precise segmentation of the shallowest tissue interface	The cGAN-based approach enhanced the visualization and segmentation of OCT images. The cascaded framework significantly outperformed existing state-of-the-art methods, demonstrating superior performance in corneal and limbal structure analysis

- **Pix2Pix-based conditional GANs (cGANs)** for predicting modeling and post-treatment image simulation
- **Deep-learning conditional GANs (DLCGANs)** for domain adaptation and multimodal translation (e.g., fundus to fluorescein angiography)
- **Low-resolution (Lr-GAN)** and **lesion-aware GAN (LA-GAN)** architectures for fundus angiographic image generation
- **CycleGANs** for ultrasound biomicroscopy (UBM) for anterior segment imaging
- **Diffusion models** as next-generation alternatives to traditional GANs for ophthalmic image synthesis

Methods

Evaluation Metrics Used for GANs

Several evaluation metrics have been used in GAN research to assess image realism and diversity in generating high-quality, realistic images in medical imaging. Assessing GANs requires a combination of metrics such as Fréchet Inception Distance (FID), Structural Similarity Index Measure (SSIM), Inception Score (IS), and computational cost metrics. These provide standardized measures for determining how accurate the GAN systems are at capturing the original image morphology and can be used to determine whether a new model presents advancements in the field.

Fréchet Inception Distance (FID) is one of the most widely used metrics to evaluate the realism and diversity of AI-generated images. It compares the distribution of feature activations of real images with those of generated images in a pre-trained Inception network. A lower FID score indicates that the generated images are closer to real images in terms of quality and diversity. For example, Jameel et al. found lower FID scores of 21.3 for in-distribution and 24.1 for out-of-distribution, indicating that the in-distribution category produced more realistic images [36]. An objective measure for the realism of synthetic images proves crucial in evaluating GANs for clinical applications like retinal image generation.

However, even though FID has been used as a reliable metric, it faces several limitations, such as inconsistency in capturing the perceptual quality of contextual reference in medical imaging, making improper normality assumptions, poor sample complexity, and inconsistent results from varying sample sizes. Jayasumana et al. proposed an alternate metric called CLIP-Maximum Mean Discrepancy (CMMD), which uses richer CLIP embeddings trained on 400 million image-text pairs containing complex scenes, as opposed to GANs that use an Inception network trained on only 1 million images, and measures an MMD with a Gaussian RBF kernel to provide more reliable data. CMMD represents an

Table 6 Comparative summary of recent studies using diffusion models for ophthalmic image analysis

Study	Model type	Application area	Key findings	Comparison to GANs	Limitations/notes
Kim et al. [52]	DDPM (U-Net-based)	Fundus photography	Trained on 1000 healthy images to generate 128 × 128 FPs; shows promise with small datasets	GANs (PGGAN) yielded sharper images with better FID (41.7 vs. 65.6)	Could not generate higher resolution (256 × 256 pixel) images due to cloud constraints
Go et al. [53]	DDPM + conditional mask	Fundus image synthesis	Conditioned generation using artery/vein masks; generated images nearly indistinguishable from real ones	Greater anatomical precision than standard GANs	Increased preprocessing complexity
Ilanchezian et al. [54]	Diffusion and robust classifier	Fundus and OCT counterfactuals	Used for generating hypothetical images for decision support; included counterfactuals	Offers newer interpretability not available through GANs	Requires robust classifier pairing
Wu et al. [55]	DDPM	OCT layer synthesis	Generated OCT images from rough sketches; improved accuracy of segmentation	Comparable performance to models trained on real data	Still computationally expensive for real-time use
Alimanov and Islam, [56]	Two-stage DDPM	Fundus image and vessel segmentation	Stage 1: vessel tree synthesis → Stage 2: fundus generation; improved vessel segmentation	Shows modular generation control superior to GANs	Two-stage approach increases training time
Ho et al. [57]	DDPM (foundational work)	General medical imaging	Introduced DDPM framework; the basis for U-Net architectures in image denoising and synthesis	More stable than GANs; no mode collapse	Slower sampling speed
Rombach et al. [58]	Latent diffusion model (LDM)	Cross-domain synthesis	LDM operates in compressed latent space, reducing computation while retaining detail	High efficiency and scalable to 512 + pixel images	Limited ophthalmic-specific applications so far

Table 7 Comparison of GANs to diffusion models in ophthalmic image synthesis

Feature	GANs	Diffusion models [52, 59]
Training stability	Often unstable	Highly stable
Mode coverage	Prone to mode collapse	Excellent
Image sharpness	High	Moderate
Training time	Faster	Slower
Data efficiency	Requires more data	Performs well with small datasets
Clinical use cases	Extensive (OCT, Fundus, FA)	Emerging (Fundus synthesis)
Key ophthalmic studies	DR-GAN, TUB-sGAN, Pix2Pix	Kim et al., 2022 (DDPM) [52]
Output control	Moderate (via conditioning)	High (via denoising steps)

improvement over FID as it does not make assumptions about the probability distribution of embeddings and is more sample-efficient. CMMD was found to be more reliable and was a more robust metric for assessing image quality upon further testing [60].

The Structural Similarity Index Measure (SSIM) assesses the image quality by comparing luminance, contrast, and structure between the real and GAN-generated synthetic images. It focuses on perceptual quality and provides an overall measure of similarity between two images to mimic human visual perception [61]. A higher SSIM score means that the synthetic image has a higher structural similarity to the real image. Zhao et al. used SSIM to compare synthetic images generated by their new Tub-sGAN against Costa et al.'s method [28, 32]. The analysis showed that their Tub-sGAN variant outperformed Costa et al.'s method for an SSIM of 0.8980 compared to 0.8716, indicating that the variant produced higher-quality synthetic images and better preserved the structural integrity of retinal vessels [28].

However, SSIM fails to capture clinically relevant finer details or higher-frequency artifacts in medical imaging. It tends to overestimate image quality in under-sampled MRI reconstructions, affecting the detection of subtle lesions. It can also produce unexpected or nonsensical results in both synthetic and real images, leading to incorrect conclusions [61, 62].

The inception score (IS) measures the performance of GANs by computing how well a pre-trained inception network can classify GAN-generated images [63]. A higher IS indicates that the generated images are both diverse and well-defined. The score is calculated by applying a pre-trained inception v3 model to the generated images and analyzing the output probabilities. A GAN model like Tub-sGAN may achieve an inception score of 8.5 when used to synthesize synthetic images of retinal conditions such as diabetic retinopathy and glaucoma, indicating that the generated images are realistic and diverse. On the other hand, a traditional GAN may score lower, indicating limited diversity in their image generation. Even though the inception score can provide a simple way of telling the scientist whether an image collection is diverse, it can be sensitive to minor

changes in the weights used by the inception model. This can lead to inconsistent evaluations and unreliable results and finds its relevance in general image generation tasks rather than medical imaging. IS also fails at capturing domain-specific variations and thus may need to be complemented with other metrics in clinical settings [64].

Computational cost metrics assess the feasibility of using GAN models in real-world medical applications. They evaluate the efficacy of GANs, including factors like model size, inference time, and processing power. If a newly developed GAN was found to be very expensive, it would be impractical to use it in specific clinical settings, especially in areas and underfunded research centers. For example, a complex GAN with a calculation time of 1.2 s per image requires more computational resources and time than a simpler model that requires 0.5 s per image. Even though the more complex GAN could produce more realistic, high-quality retinal images in ophthalmology, the cost of running the system and waiting for results may outweigh the benefits in high-flow workstations where faster computing is needed. However, computational cost metrics fail to consider the environmental implications of model training, such as carbon waste, and are thus frequently excluded from evaluations [65].

Generative Adversarial Networks for Image Synthesis

There is abundant literature proving that GANs can have real-world applications in the field of ophthalmology, specifically in enhancing training data sets, refining diagnostic algorithms, and improving disease detection. Choi et al. highlighted how GANs can be used for automated disease detection. This study exemplifies how synthetic data can supplement limited clinical datasets for rare ocular surface disorders like dry eye, enabling more robust model training while minimizing biases associated with limited real-world samples [66]. This technology can help mitigate biases associated with real-world sample scarcity, ultimately enabling more robust model training in the face of rare ocular disorders.

Conditional GANs and CycleGANs for Disease-Specific Fundus Image Synthesis

The use of GANs in fundus image synthesis is an emerging technology with significant implications for medical imaging, particularly in training diagnostic models and generating realistic retinal images without extensive reliance on real patient data [4, 27–29]. One prominent method involves conditional GANs, which utilize disease-specific labels to guide the image-generation process. This ensures that the generated images accurately reflect specific retinal pathologies, such as diabetic retinopathy or glaucoma, allowing these synthetic datasets to be used effectively for training machine learning models without requiring large volumes of patient data [30].

Another notable approach is image-to-image translation, exemplified by CycleGANs, which facilitates the transformation of images from one domain to another (e.g., from healthy retinal images to images of retinas with specific diseases) without requiring paired data. This allows for the flexible generation of diverse pathological images, filling critical gaps in datasets where healthy-to-disease image pairs may not exist [31].

Dual-phase GANs, Tub-GANs, and Tub-sGANs for Healthy Fundus Image Generation

In the generation of healthy fundus images, Costa et al. pioneered an adversarial learning system for producing high-quality synthetic fundus photographs. Their approach utilizes a pre-trained vascular structure model, leveraging a dataset of 614 standard fundus images to generate realistic synthetic retinal images that lack significant inter-sample variability due to the absence of disease-related features [32]. Building on this, Guibas et al. introduced a dual-phase GAN process where synthetic retinal vasculature is generated and then translated into realistic fundus images. These generated images were subsequently used to train a U-Net segmentation model, achieving an *F1* score comparable to models trained on authentic clinical data [33].

Zhao et al. proposed a GAN framework capable of training on a minimal dataset of only 10 to 20 real images, introducing two specialized models: tubular-GAN (Tub-GAN) and tubular style-transfer GAN (Tub-sGAN). These algorithms were designed to enhance the accuracy of segmentation tasks by incorporating style transfer mechanisms within the GAN architecture. Tub-sGAN, in particular, represents the first attempt to combine style transfer with GAN-based retinal image synthesis, significantly improving the realism and diversity of generated images [28].

Tub-GAN was developed to generate realistic retinal and neuronal images from tubular structure annotations, such as vessel morphology, using a patch-based convolutional

neural network (CNN) with a 27×27 patch size as input. TUB-sGAN further refines the process of diverse image generation from a single tubular structure annotation made by Tub-GAN, improving the quality and diversity of the synthesized images. The study found that adding synthetic images generated by Tub-GAN to real images improved segmentation performance by 1.18% compared to training only on real images [28]. This system is similar to an approach pioneered by Gatys et al., who used image style transfer to modify an image's texture to match a reference style [67]. This team applied CNNs to separate and capture an image's content and style. It demonstrated that these systems worked together to generate several realistic synthetic images from a single tubular structure annotation while preserving the original image morphology [67]. This marked a great advancement in the field of GAN-based retinal image synthesis.

Using Pix2pix in Approaches for Glaucoma-Specific Image Synthesis

Yu et al. introduced a novel GAN-based framework incorporating a multi-channel and multi-landmark (MCML) algorithm for generating glaucoma-specific fundus images. By focusing on critical retinal landmarks such as the optic disc (OD) and optic cup (OC), this system accurately simulates the cup-to-disc ratio (CDR), an essential metric in glaucoma diagnosis [68]. The MCML approach substantially expands the volume of available glaucoma-related images, addressing the chronic data scarcity in computer-aided diagnosis (CAD) systems for glaucoma [34].

In comparative studies, the Pix2pix model integrated with a ResU-Net generator demonstrated superior performance in generating high-resolution synthetic fundus images, surpassing traditional GAN architectures in terms of peak signal-to-noise ratio (PSNR) and Structural Similarity Index (SSIM). The combination of Pix2pix, a ResU-Net generator, and the MCML approach enabled the creation of high-quality, disease-specific fundus images, making it an invaluable tool for augmenting training datasets in CAD systems [34]. This system's ability to generate lifelike, high-resolution images from small datasets represents a significant advancement in AI-driven ophthalmic diagnostics, particularly in glaucoma management.

Pix2pix-based cGANs for Predictive Modeling and Post-Intervention Assessment

GANs are also increasingly used in predictive modeling, particularly in forecasting post-intervention outcomes. Lee et al. developed a cGAN model to predict post-treatment OCT images for patients with neovascular age-related macular degeneration (AMD). Using 14,183 paired images, the model accurately predicted changes in key indicators such as

intraretinal and subretinal fluid, pigment epithelial detachment, and subretinal hyperreflective material. Incorporating baseline angiography data further improved the model's performance, achieving accuracies between 80.7% and 96.3% and specificity rates above 97% in predicting these post-treatment features [46].

Similarly, Liu et al. applied a Pix2pix-based cGAN to predict the response of neovascular AMD to anti-vascular endothelial growth factor (anti-VEGF) therapy. Using 476 paired pre- and post-treatment OCT images, the model generated synthetic post-treatment images clinically interpretable in 92% of cases, with only 26–30% identified as synthetic by experts [47]. These predictive capabilities underscore the potential for GANs to assist in treatment planning and outcome assessment in ophthalmic diseases.

Keratoconus is a corneal condition characterized by progressive thinning and bulging of the cornea, resulting in irregular astigmatism and vision loss. Various medications can lead to abnormal corneal curvature, including cationic amphiphilic drugs such as amiodarone, antimalarials (chloroquine, primaquine), tamoxifen, and NSAIDs [69]. Early detection is crucial for timely intervention and improved outcomes; however, the most reliable and accessible diagnostic tools, such as Scheimpflug and anterior segment optical coherence tomography, often necessitate large, well-annotated datasets for deep machine learning models. These devices can evaluate both the front and back surfaces of the cornea, create a corneal thickness map, and reconstruct the anterior surface for detailed analysis.

To address this issue, Abdelmotaal et al. employed a pix2pix conditional GAN (cGAN) to generate high-resolution, synthetic color-coded corneal tomography images from a Scheimpflug camera to train a deep convolutional neural network (DCNN) for keratoconus detection [70]. Unlike traditional GANs, which generate images without specific input constraints, the new pix2pix cGAN operates based on the mapping between paired input and output images, making it practical for tasks like image enhancement, segmentation, and synthesis while allowing it to learn the mapping between lower-quality or partial corneal images and their high-quality, color-coded counterparts [71]. This process ensures that the generated synthetic images preserve key anatomical features while enhancing resolution and diagnostic value. This type of cGAN can yield less precise outputs as it is not trained on paired data [71].

Abdelmotaal et al. found that incorporating GAN-generated synthetic images with real images improved the performance and model training of DCNNs, overcoming challenges such as limited datasets and class imbalance in diagnostic models. By augmenting datasets with GAN-generated synthetic images, the model became more robust and succeeded in the early detection of keratoconus that lower-quality inputs might not represent. However, the reliance on

pix2pix on well-annotated datasets presents a limitation for keratoconus, where dataset scarcity is a common barrier to model training. Overall, the study demonstrated that integrating synthetic OCT images into pix2pix cGAN models can significantly enhance DCNN model performance for keratoconus detection, highlighting the potential impact of GANs in advancing ophthalmic imaging [71].

Literature has also applied GANs to meibomian gland dysfunction (MGD) analysis. MGD is characterized by gland dropout and abnormal secretions. Traditional evaluation methods depend on subjective measures, which can introduce significant diagnosis variability. Recent studies have utilized CycleGANs and conditional GANs to improve automated MGD detection from infrared (IR) meibography images. Khan et al. applied a cGAN-based approach to segment and quantify meibomian gland atrophy, achieving significantly better model performance than traditional methods like adaptive thresholding and U-Net segmentation. The GAN model exhibited a high correlation ($r = 0.96$, $p < 0.001$) between automated and manual analyses, enhancing diagnostic accuracy while reducing observer-dependent variability. Thus, GAN-generated synthetic meibography images can facilitate deep machine-learning training to improve performance across diverse patient populations [39].

DR-GANs for Synthesizing Fundus Images with Disease Features

GANs have been increasingly utilized to generate fundus images depicting ocular conditions such as age-related macular degeneration (AMD), glaucoma, diabetic retinopathy (DR), and retinopathy of prematurity (ROP). A notable application is the DR-GAN, introduced by Zhou et al., which addresses the challenge of detecting advanced stages of DR—where data scarcity often hinders model training—by generating high-quality fundus images annotated with specific lesion characteristics [30]. DR severity, traditionally categorized into five levels, requires diverse training datasets for accurate classification. Standard data augmentation techniques fail to capture the necessary variability, but DR-GAN overcomes this by synthesizing customized images with adaptive grading based on lesion masks and structural elements [30]. By leveraging multiscale spatial and channel attention (SCA) mechanisms, DR-GAN enhances the precision of fine details, producing highly realistic 1280×1280 fundus images that support accurate grading of DR [30]. Testing on EyePACS and FGADR datasets confirmed the system's ability to generate fundus images that mirror real-world data, significantly improving the availability of high-resolution, disease-graded images for model training [30].

Zhou et al. further explored the role of GANs in data augmentation for DR image analysis, evaluating whether DR-GAN could mitigate data imbalance issues in grading

models. The study tested two experimental setups: one using only DR-GAN synthesized data and another combining real and synthetic images for model training. Models were assessed using classification accuracy and kappa metrics, revealing significant improvements—accuracy increased by 1.75% and kappa by 1.87%. Additionally, true positive rates per class showed enhancements across all categories when synthesized samples were included [30]. These findings underscore the effectiveness of GAN-based data augmentation in addressing dataset limitations for rare ocular diseases, ultimately advancing ophthalmic AI applications and improving patient treatment options.

In a separate study by Burlina et al., GANs were trained on 133,821 color fundus photographs from 4613 participants in the Age-Related Eye Disease Study (AREDS), generating synthetic fundus images both with and without AMD features [72]. Remarkably, the diagnostic accuracies of retinal specialists when evaluating real versus synthetic images were nearly identical: for the first specialist, accuracy was 84.54% with real images and 84.12% with synthetic images, with error margins of 4.06% and 4.16%, respectively [72]. For the second specialist, accuracies were 89.47% for real images and 89.19% for synthetic ones, with error margins of 3.45% and 3.54% [72]. Notably, the specialists struggled to differentiate between real and synthetic images, achieving accuracy rates of 59.50% and 53.67% [72]. Furthermore, convolutional neural networks (CNNs) trained on synthetic datasets achieved an area under the curve (AUC) of 0.9235, closely matching the AUC of 0.9706 for models trained on real data [72]. These results demonstrate the high degree of realism in GAN-generated images and their viability in training deep-learning models for AMD diagnosis [72].

Diaz-Pinto et al. further advanced GAN-based image synthesis by developing a framework for generating synthetic retinal images to assist in glaucoma detection [35]. Their semi-supervised approach utilized a modestly sized labeled dataset of glaucoma images alongside a much larger unlabeled dataset, totaling 86,926 publicly available images [35]. The GAN-generated synthetic images were validated using t-SNE visualization to assess anatomical accuracy, focusing particularly on the alignment of pixel distributions with optic disc structures. This method allowed the model to distinguish between glaucomatous and healthy conditions with an AUC of 0.9017 [35]. The ability to autonomously generate and label synthetic images presents a critical advantage in addressing data scarcity in glaucoma detection models [35].

GANs in Optical Coherence Tomography (OCT)

Generating realistic OCT images presents unique challenges due to the detailed structural and textural information these scans contain. Traditional GAN architectures often struggle

to capture these complexities, leading to synthetic images that may lack the clinical accuracy necessary for use in diagnostic settings. To address these challenges, researchers have developed novel GAN architectures and training methods specifically designed for OCT data, yielding promising results across various applications, including image synthesis, enhancement, post-intervention prediction, and domain transfer [40].

PGGANs, DCGANs, and StyleGANs for Image Synthesis in OCT

GANs have shown substantial success in generating synthetic OCT images, which can be used for model training and data augmentation. In one notable study, Zheng et al. generated high-resolution OCT images from a dataset containing tens of thousands of cases, including choroidal neovascularization, diabetic macular edema, drusen, and normal eyes [40]. Two deep-learning models were trained to distinguish urgent from non-urgent cases using these images. When tested on both local and clinical datasets, the models achieved AUC scores of 0.99 and 0.98 for local data and 0.94 and 0.90 for clinical data, respectively [40]. Retinal specialists tasked with differentiating real from synthetic OCT images had difficulty achieving accuracy rates above 60%, highlighting the realism of the synthetic images [40].

Similarly, Kumar et al. employed progressive GANs (PGGANs) to generate OCT images for glaucoma detection. The synthetic images were used to train deep-learning networks, which achieved AUC scores comparable to networks trained on real data. Specifically, the networks trained on synthetic data achieved AUCs of 0.97 on internal datasets and 0.90 on external datasets, compared to 0.96 and 0.84 for real data-trained networks [41]. These results demonstrate that GAN-generated OCT images can closely mimic real data, making them valuable for training and diagnostic tasks in clinical environments.

Liu et al. also published a study on generating synthetic retinal images to demonstrate the use of GANs in mimicking images exhibiting age-related macular degeneration (AMD) symptoms [47]. The researchers employed deep convolutional GANs (DCGANs), which build on traditional GAN architecture by replacing its components with fully convolutional layers, and Wasserstein GAN, which introduces the Wasserstein distance and provides more stable measures of the difference between two probability distributions, to produce images with detailed pathological features. These synthetic images were utilized to train classifiers, improving accuracy in distinguishing between different AMD subtypes. The study found that the generated images contained sufficient pathological details to facilitate ophthalmologists' disease classification and feature extraction tasks [73].

Another type of GAN, a StyleGAN, was used to create high-quality synthetic OCT B-scans from patients with and without AMD that even retinal experts could not distinguish from real images and was used to create high-resolution retinal images for use in diagnostics and augmenting data sets [74, 75]. StyleGAN differs from traditional GANs by introducing a novel “style”-based architecture that allows for greater control over the generated images. Researchers achieved this through a mapping network that transforms the input latent code into a style vector, modulating the generator’s intermediate layers. This design manipulates specific features such as pose, expression, and lighting in generated images [76].

DN-GAN and CycleGANs for Image Enhancement and Noise Reduction

Beyond image synthesis, GANs have also been widely applied to enhance the quality of OCT images. Conditional GANs (cGANs) have proven particularly effective in both supervised and unsupervised settings for tasks such as speckle noise reduction and resolution enhancement. Huang et al. introduced the SDSR-OCT model, which simultaneously performs denoising and super-resolution, significantly improving image quality compared to traditional techniques [10]. Similarly, Chen et al. developed a denoising GAN (DN-GAN), a denoising framework that maps high-noise images to low-noise counterparts through end-to-end training, reducing speckle noise and improving diagnostic accuracy in OCT images [42].

To further enhance the utility of GANs in OCT, Ma et al. designed a cGAN framework incorporating an edge loss function that emphasizes the preservation of anatomical boundaries during denoising [43]. This model excelled in generalizing across different retinal OCT datasets, with a denoising speed of 0.22 s per B-scan, highlighting its potential for real-time clinical use [43]. Other studies, such as DeshadowGAN and OCT-GAN, focus on removing shadow artifacts caused by blood vessels or calcifications, further refining the quality of OCT scans for diagnostic purposes [44, 45].

Manakov et al. introduced a new method for denoising retinal optical coherence tomography (OCT) images by treating noise as a domain shift and employing a modified version of CycleGAN, introduced by Zhu et al., for unpaired image translation [43, 77]. The CycleGAN is uniquely able to facilitate the transformation of images across two different domains, such as noisy and clean OCT images, without requiring paired data sets for training. This is particularly helpful in the medical setting for rare ocular disorders, where it can be challenging to obtain paired data.

The CycleGAN does so by combining two GANs into a unified two-way autoencoder. Each GAN transforms images

from one domain to another. One generator translates images from the noisy domain of raw OCT images into a cleaner version. In contrast, the other generator reverses the process to ensure the network can map pixels to their counterparts. The discriminator in each GAN can then evaluate each transformed image to determine how realistic they are in improving machine learning. This modified GAN also uses a built-in “cycle-consistency” loss to ensure that any synthetic images still accurately portray the original image after transformation across domains, allowing for the data set to maintain integrity and accuracy. This allows bidirectional image translation without requiring paired data training, a standard limitation of other translation techniques, and is useful when labeled data is unavailable. The team found that this novel CycleGAN effectively reduced noise in OCT images, outperforming established methods in both quantitative measurements and qualitative evaluations by ophthalmologists. The model also demonstrated the ability to differentiate between subtle changes in noise levels and distinguish between retinal layers and other distinct regions within the OCT images. This model has also been used to translate fundus photographs to fluorescein angiography images, aiding in the development of multi-modal imaging approaches [43, 77].

Deep-Learning cGANs for Domain Transfer and Multimodal Imaging

Another important application of GANs in OCT involves domain transfer, where GANs are used to translate between different imaging modalities or devices. For example, Tavakkoli et al. developed a deep-learning conditional GAN (DLCGAN) model to generate fluorescein angiography (FA) images from fundus photographs, leveraging a dataset of both normal and diabetic retinopathy patients. The GAN-generated angiograms were anatomically accurate and outperformed state-of-the-art generative models in producing realistic images [51]. Such domain transfer techniques could allow clinicians to generate multimodal images without requiring additional invasive procedures, significantly expanding the utility of OCT imaging.

Lr- and LA-GANs in Fundus Fluorescein Angiography

Fundus fluorescein angiography (FFA) is a diagnostic imaging technique that involves injecting fluorescein dye into the bloodstream to visualize the vascular structures of the retina and choroid [78]. Although effective, FFA is invasive and can be uncomfortable for patients, prompting the development of methods to generate these images non-invasively using GANs.

Li et al. addressed this challenge by developing a dual-GAN system, consisting of a low-resolution GAN (LrGAN)

to capture global intensity details of the FFA images and a high-resolution GAN (HrGAN) to generate high-definition patches [79]. This approach combined both supervised and unsupervised learning techniques to achieve a balance between fine detail generation and global image consistency. The model significantly outperformed traditional methods, achieving a structural similarity index (SSIM) of 0.7126, a normalized cross-correlation (NCC) of 0.6799, and a peak signal-to-noise ratio (PSNR) of 15.77 [79]. The results demonstrated the potential of this multi-frame, high-resolution approach for generating high-quality FFA images directly from retinal fundus photographs, offering a less invasive alternative for retinal imaging.

In another study, Huang et al. developed lesion-aware GANs (LA-GANs), an adversarial multi-task learning model designed to enhance lesion characteristics in FFA images generated from color fundus photographs [80]. LA-GAN introduced a region-level adversarial loss, which more accurately measured discrepancies between corresponding regions of varying shapes. This model used a multi-task network architecture specifically designed to handle high-resolution image processing, enabling it to outperform other image translation techniques. The SSIM index reached 0.536, and the Learned Perceptual Image Patch Similarity (LPIPS) was 0.312 [80].

More importantly, LA-GAN also significantly improved lesion-region segmentation performance. The mean Dice coefficient increased from 0.714 to 0.797, while the mean accuracy rose from 0.873 to 0.905 [80]. These results underscore the model's superiority in lesion identification and segmentation, surpassing existing FFA generation methods and demonstrating the potential for broader application in both image translation and lesion detection tasks [80]. LA-GAN's performance in these tasks indicates its potential for improving clinical workflows by enhancing diagnostic precision in retinal angiography.

CycleGANs in Ultrasound Biomicroscopy (UBM)

Ye et al. explored the use of CycleGAN to transform swept-source anterior segment optical coherence tomography (SS-ASOCT) images into synthesized ultrasound biomicroscopy (UBM) images [81]. The study aimed to evaluate the model's ability to generate high-quality synthetic UBM images, assessing both the visual fidelity of the images and their diagnostic utility. Glaucoma specialists were tasked with comparing the synthetic images to real SS-ASOCT images, and the model's performance was further evaluated through measurements of key anterior chamber and iridociliary parameters [81].

The results showed that the synthetic UBM images closely matched the real SS-ASOCT images in terms of visual quality, with a high level of consistency in measurements

between the real and synthetic datasets. The Bland–Altman analysis confirmed a strong agreement in anterior segment measurements, indicating that the synthetic UBM images could be effectively used for diagnostic purposes in clinical settings [81]. Moreover, the synthetic images achieved Fréchet Inception Distance (FID) scores of 21.3 for in-distribution datasets and 24.1 for out-of-distribution datasets, reflecting high image quality and realistic synthesis, even when the input data was outside the model's primary training domain [81].

These findings demonstrate the potential of GANs, particularly CycleGAN, in generating realistic UBM images from SS-ASOCT data, reducing the need for direct UBM imaging and offering a non-invasive alternative for anterior segment analysis in glaucoma diagnostics.

Diffusion Models as Emerging GAN Alternative Architectures in Ophthalmic Imaging

Diffusion models have emerged as a promising alternative to generative adversarial networks (GANs) for synthesizing ophthalmic images, offering advantages in training stability and data distribution coverage. Unlike GANs, which can suffer from training instability and mode collapse, diffusion models iteratively refine random noise into high-fidelity images through a reverse denoising process, allowing for better coverage of data distributions and more stable convergence [59].

In a feasibility study, Kim et al. explored the use of denoising diffusion probabilistic models (DDPMs) to generate fundus photographs from limited datasets. They trained a DDPM with a U-Net architecture on 1000 health retinal images, successfully producing 128×128 pixel synthetic fundus photographs (FPs) [52]. The U-Net architecture is the most general form of the generative diffusion mode and was used to generate realistic FPs from random noise. However, they could not generate higher resolution, trained 256×256 pixel images, because of limited computation capacity on the cloud platform. Results showed that the progressive growing generative adversarial network (PGGAN) made more sharpened images than the DDPM in terms of retinal vessels and optic discs, as indicated by a lower Frechet inception distance (FID) of 41.761 compared to that of DDPM with an FID of 65.605. The study concluded that even though DDPMs can generate realistic FP images from small datasets, they were limited in training stability and image quality compared to GANs like PGGAN [52].

Subsequent studies have built upon these findings. Go et al. introduced a diffusion model that generates anatomically accurate retinal fundus images by conditioning on artery/vein masks, resulting in high-quality images that were difficult for clinical experts to distinguish from real ones [53]. Similarly, Ilanchezian et al. combined diffusion

models with adversarially robust classifiers to generate realistic counterfactual retinal fundus and OCT images, aiding in clinical decision-making by illustrating hypothetical disease scenarios [54]. Wu et al. employed DDPMs to synthesize retinal OCT images from rough layer sketches, enhancing layer segmentation accuracy and demonstrating that models trained solely on synthetic images could perform comparably to those trained on real data [55]. Additionally, Alimanov and Islam developed a two-stage DDPM that first generates vessel trees and then fundus images conditioned on these structures, achieving high-quality synthetic images and improving vessel segmentation tasks [56, 59, 82].

These advancements suggest that diffusion models can complement GANs in ophthalmic image synthesis, particularly in scenarios with limited training data or when structural accuracy is paramount. The ability of diffusion models to generate diverse and anatomically consistent images makes them valuable for data augmentation, training of diagnostic models, and potentially for simulating disease progression [59, 82]. As computational resources become more accessible and model architectures continue to evolve, diffusion models are poised to play an increasingly significant role in ophthalmic image analysis and synthesis [59, 82].

Challenges, Gaps, and Future Directions

Technical Limitations of GANs in Ophthalmology

While GANs have the potential to impact the field of ophthalmology, there are many hurdles that must be overcome before they can be implemented in real-world applications. Despite the transformative potential of generative AI, current models exhibit critical limitations across key domains, including accuracy, reliability, validity, reproducibility, and generalizability, particularly in their integration into clinical workflows [83]. Zhang et al. underscore that there are many strict requirements that AI technologies must meet to be a viable and practical option in clinical decision-making, emphasizing the challenges of embedding these systems within existing healthcare infrastructure without jeopardizing patient outcomes [83]. However, it is still debatable whether AI-generated data can seamlessly integrate with real-world patient cases without compromising outcomes.

In practice, GANs are prone to training instability and mode collapse, sensitivity to hyperparameters, and issues capturing fine-grained details necessary in the diagnostic process. GAN training requires careful balancing in the weights used by the generator and discriminator networks. Thus, imbalances can lead to overfitting, vanishing gradients, or failure to converge [84]. Mode collapse refers to the limited production of outputs even in the face of diverse inputs, which reduces the use of synthetic datasets [85]. Hyperparameter choices, such as learning rate, batch size,

and network architecture, require extensive trial and error by generating GAN images. A lack of standardized metrics and practices makes adoption difficult for everyday clinicians who may not be familiar with AI [86].

Additionally, GANs are limited in capturing subtle pathological findings, particularly in complex OCT scans, where microscopic abnormalities are essential in accurate diagnoses [87]. Computational cost metrics for training GANs also raise concerns about the feasibility of applying them in the real world. If it costs too much to use a GAN in a resource-limited environment, it could limit more widespread adoption.

Validation, Bias, and Image Quality Concerns

As the use of GANs expands, the emergence of deepfake technology and AI-generated medical content raises concerns regarding image authenticity, verification, and patient safety. The possibility of AI systems generating false or misleading medical images necessitates stringent regulatory oversight to maintain the integrity of clinical decision-making [88]. Without robust validation methods, such as the FID and Inception Score, there is a risk of erroneous diagnoses and false clinical interpretations stemming from unreliable synthetic data [88–90].

In ophthalmology, these risks are magnified due to the sensitive nature of biometric data and the high stakes associated with vision-related diagnosis. The lack of standardized validation metrics and concerns over data privacy and potential algorithmic biases create ethical challenges in model development and deployment. Liu et al. highlight the dual challenge of mitigating bias while maintaining strict data protection protocols, advocating for fairness and transparency as the central tenets in developing and implementing AI systems in medical contexts [91]. Oniani et al. further elaborate on these concerns, highlighting equity, autonomy, informed consent, and privacy as critical ethical pillars that must guide the deployment of AI-driven technologies in clinical practice [27, 89].

Several studies have also raised concerns about the risk of synthetic images contributing to misdiagnosis [89, 90]. To address the technical limitations of current deepfake detection methods—which often overfit specific artifacts and fail to generalize across GAN models—Liu et al. proposed a novel multi-view image completion framework. This system reconstructs missing image components from multiple perspectives to learn frequency-irrelevant features and better capture the distributional characteristics of authentic medical images. This model incorporates intra- and inter-view learning strategies, a low-pass residual-guided attention module, and multiscale feature concatenation to enhance representation robustness against unknown synthetic manipulations [91].

Ethical, Privacy, and Legal Considerations

The adoption of GANs in clinical environments brings new bioethical and legal questions. These include algorithmic bias risks, misuse of synthetic data, concerns about regulatory loopholes for anonymized yet clinically significant synthetic images, and the unregulated use of AI-generated content that closely mimics real patient anatomy [92, 93]. For example, synthetic data generated by GANs may not be adequately covered by standard data protection legislation, raising the risk that such data could be used to circumvent privacy laws or be subject to misuse in ways that affect patient trust and safety [92, 93]. Algorithmic bias remains a significant concern, as the quality and diversity of training data directly influence fairness and accuracy in clinical applications; regular auditing and human oversight are recommended to mitigate these risks [92–94]. As a result, there is a growing need to clarify whether GAN-generated images fall under the same legal protections as patient-derived data.

Liu et al. emphasize the key role of regulatory bodies such as the FDA and EMA in establishing clear guidelines for validating and approving AI-generated medical data to ensure it meets clinical safety standards [92, 95, 96]. Strict frameworks are essential for technical validation and ensuring patient trust, informed consent, and the ethical use of synthetic data within clinical practice. The FDA's Ophthalmology Program underscores the need for standard methods to evaluate system performance and aid clinical translation of innovative devices, including those using AI and GANs [92, 96–98]. The FDA and EMA are actively adapting their approval pathways—such as 510(k), de novo, and the more rigorous premarket approval (PMA)—to accommodate AI-enabled technologies, with GAN-based systems often falling into higher-risk classifications due to their complexity and potential clinical impact [98–101].

These regulatory frameworks also emphasize key principles such as explainability, auditability, and data governance while responding to global calls for harmonization through initiatives like the Accelerating Clinical Trials in the EU (ACT EU) [96, 102]. Moreover, organizations like the U.S. National Institute of Standards and Technology (NIST) are developing standards for managing the risks associated with sensitive biometric data and ensuring robust data security [92, 96]. These developments indicate that while GANs hold transformative potential for clinical ophthalmology, their successful implementation will require interdisciplinary collaboration, evolving regulatory policy, and adherence to rigorous validation and ethical standards.

Educational institutions have begun incorporating GAN-generated datasets into medical education, citing privacy benefits and fidelity to real-world anatomy. Still, policies must clarify whether synthetic data falls under the same regulatory purview as patient-derived data since the legal status

of synthetic images—especially those that closely mimic real patient anatomy—remains ambiguous under current privacy frameworks [92–94, 103].

Potential Solutions to the Limitations and Ethical Considerations Associated with Generative AI Use

Federated Learning and Secure Model Training

To address privacy and generalizability concerns, researchers are turning to federated learning (FL)—a collaborative training approach where a central server controls model updates across multiple sites without sharing raw patient data and poses a worst-case privacy risk. Ziller et al. argued that using federated learning with central servers can ameliorate the risk of malicious attacks manipulating training parameters and architectures that could affect privacy risks by precisely reconstructing input data. This system would also allow hospitals to train a shared AI model without directly sharing patient data, ensuring that raw data stays on local servers and decentralizing sensitive information [104]. This system may become impractical if the sensitive data inferred by the center server is not adequately protected, and model updates would be transmitted across all institutions using the same system. They addressed this by incorporating differential privacy (DP) and secure aggregation techniques like homomorphic encryption and secure multi-party computation, thereby decentralizing sensitive information and enhancing model security and utility [104]. Integrating federated learning with generative adversarial networks (GANs) enhances privacy and data utility by enabling collaborative model training without exposing sensitive information. Recent applications of federated learning in healthcare and medical imaging demonstrate that this approach improves model generalizability while preserving privacy and facilitating cross-institutional validation [105, 106].

Explainable and Low-Resource AI for Clinical Trust and Access

Similarly, explainable GAN (X-GAN) frameworks are being developed to increase transparency by identifying which features in the input contribute to the generated output, thereby aiding clinical interpretability and trust. Recent research highlights the importance of explainability in medical AI, noting that interpretability-based models, attention-based mechanisms, and attribution maps in explainable AI techniques can help clinicians understand and trust complex model predictions by clarifying the contribution of input features to outputs [107–111].

Studies have also shown that incorporating clinical validation studies and expert-driven assessments beyond traditional metrics like FID and SSIM can provide a more

comprehensive evaluation of synthetic medical images [107]. In addition, due to privacy issues surrounding patient files, regulatory bodies must collaborate with AI researchers and clinicians to establish guidelines that ensure the safe and ethical deployment of GANs in ophthalmology. By addressing these concerns, GANs can be more effectively integrated into clinical workflows, enhancing diagnostic capabilities while maintaining ethical integrity and patient safety.

Complementing explainability is the need for accessibility. Low-resource deployment strategies, such as lightweight GAN architectures and model compression, are also being explored to extend the reach of generative models to mobile platforms, point-of-care settings, and underserved areas where computational capacity is limited [109].

Standardization and Translating GANs from Research into Clinical Practice

A critical tool in enabling the clinical adoption of GANs in the standardizing evaluation and reporting practices. Initiatives like STARD-AI, CLAIM, TRIPOD-AI, and CONSORT-AI aim to improve AI diagnostic studies' transparency, completeness, and reproducibility [112, 113]. These frameworks also emphasize human oversight and interdisciplinary collaboration, ensuring stakeholders can appraise AI models' quality and performance in healthcare settings [112, 113].

Translating generative adversarial networks (GANs) from research to clinical practice in ophthalmology involves addressing several key areas, including integrating into clinical workflows, ensuring generalizability across diverse datasets and imaging devices, and developing multimodal systems that can be incorporated into electronic health records (EHRs) and decision-support tools.

Clinical Workflow Integration

GAN-based models like DeshadowGAN can effectively remove blood vessel shadows in OCT images of the optic nerve head, improving image quality and potentially aiding diagnosis. Similarly, multiscale denoising GANs have been developed to reduce speckle noise in OCT images, enhancing the clarity of retinal structures. These advancements suggest that integrating GAN-based denoising into OCT devices could streamline clinical workflows by providing clearer images for immediate interpretation [58].

Generalizability Across Datasets and Devices

A significant challenge in deploying GANs clinically is ensuring their performance across diverse datasets and imaging devices. Differences in image characteristics between devices can affect the accuracy of diagnostic models. To

address this, researchers have employed image-to-image translation GANs (e.g., CycleGANs) to harmonize OCT images from different devices, such as translating images from Cirrus to Spectralis systems, thereby improving the generalizability of segmentation and diagnostic models [114–116]. For example, GANSeg demonstrated robust unsupervised cross-domain learning, enabling accurate segmentation of OCT images from devices not present in the original training data [114, 115]. Such domain adaptation techniques are crucial for developing robust models applicable in varied clinical contexts, from tertiary hospitals to resource-limited settings [114, 117].

Multimodal Systems and EHR Integration

GANs also facilitate the development of multimodal imaging systems that can be integrated into EHRs and decision-support tools. For example, conditional GANs have been used to synthesize fluorescein angiography images from fundus photographs, providing non-invasive alternatives for visualizing retinal vasculature [51, 58]. These synthetic images can be incorporated into patient records, enhancing diagnostic capabilities without additional invasive procedures. GAN-based fusion models integrate imaging and EHR data and improve predictive performance for conditions like glaucoma, supporting more comprehensive patient assessments and informed clinical decision-making [103].

However, real-world implementation faces barriers such as insufficient interoperability between GAN-enhanced outputs and existing clinical systems and a lack of standardized protocols for incorporating synthetic images into clinical workflows. Overcoming these barriers will require technical refinement and adaptation of health IT policy.

In conclusion, the adoption of GANs in ophthalmology from research to clinical practice depends on their smooth integration into clinical workflows, versatility across different datasets and devices, adherence to regulatory requirements, and integration into multimodal systems connected to electronic health records. Realizing the full potential of GANs in improving ophthalmic care requires ongoing research and cooperation between technologists, clinicians, and regulatory agencies [92–94, 103, 112, 113].

Conclusions

The application of GANs in ophthalmology represents a shift in how clinical decision-making, the use of diagnostic tools, treatment planning, and patient care can be greatly enhanced by using AI. GANs have enabled significant advancements in training diagnostic algorithms by synthesizing high-resolution, disease-specific ophthalmic images, particularly in scenarios where access to

large, annotated datasets is limited. The ability of GANs to generate realistic images—whether for retinal, OCT, or fluorescein angiography—has been shown to improve diagnostic accuracy for diseases like diabetic retinopathy, glaucoma, and age-related macular degeneration [4, 26, 30, 47, 59, 81, 103, 118].

However, several challenges persist in the clinical integration of GAN-generated images. Significant obstacles to broad adoption include the lack of standardized evaluation metrics for synthetic images, intrinsic biases in medical imaging data, and the requirement for more interpretable deep-learning models. When training data is scarce, methods like transfer learning may help enhance GAN performance. Additionally, image enhancement techniques like super-resolution and denoising have shown promise in raising the clinical fidelity of synthetic images [4, 26, 30, 47, 59, 81, 103, 118].

Future research must focus on refining GAN architectures to further boost the realism and utility of synthetic images while addressing ethical considerations and regulatory challenges. A multi-disciplinary approach—combining expertise from ophthalmology, AI, and regulatory agencies—will be essential in translating these advancements into clinical practice. By fostering cross-disciplinary collaboration and open data sharing, GANs can revolutionize ophthalmic diagnostics, particularly in resource-constrained settings, thereby improving patient outcomes globally [4, 26, 30, 47, 59, 81, 103, 118]. By embracing federated learning, explainable AI, low-resource deployment, and international standardization, the field can move toward safe and equitable integration of generative technologies into clinical workflows [104–106].

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Declarations

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